Sahaj Bhadja

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Summary

Software engineer experienced in **Python, C++, and TypeScript**, building backend services, product interfaces, data pipelines, and computer vision systems from design through testing and deployment.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Built Python FastAPI data pipelines across PostgreSQL and cloud storage, processing 5,000+ monthly jobs with validation, reconciliation, and CI/CD-backed releases, achieving 99%+ reliability in production environments at scale.
- Architected a multi-tenant platform on Azure SQL and PostgreSQL for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through query optimization and implementation-level tuning.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Designed React interfaces and Python Flask backends for an AI grading product serving 150+ instructors, processing 10,000+ daily requests with MySQL persistence and stable API performance at scale.
- Integrated computer vision models including GPT-4o Vision and Claude Vision to evaluate 10,000+ handwritten submissions, adding documented fallbacks and automated tests while reducing feedback turnaround time 60% at scale.

USC AD Labs **Los Angeles**
ML Research Assistant *January 2025 – May 2025*
- Built a GPT-based Decision Transformer for offline reinforcement learning in autonomous driving, improving long-horizon trajectory planning 18% in simulation while processing 100,000+ trajectory logs end to end.
- Architected a spatial-temporal encoder integrating HD-map geometry covering road layout and lane direction with ego-agent past trajectories via multi-head attention, improving closed-loop planning stability 15% at scale.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Built Jenkins CI/CD tooling and 20+ unit and integration tests for Java and Python microservices, increasing pipeline efficiency 68% while collaborating with Microsoft and GitHub stakeholders at scale.
- Optimized large-scale API integrations using Python, AWS Glue, and S3, improving response times 30% for low-bandwidth users while reducing technical debt 20% across production ETL workflows end to end.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Developed a React and Node.js resource management platform with REST APIs and eight microservices, syncing two million records while reducing resource wastage 42% and increasing conversions 15% at scale.
- Shipped product features end to end in a fast-paced environment, using Google Analytics and Sentry monitoring to debug production issues while reducing deployment time 50% through Jenkins CI/CD.

Skills

Languages: Python, C++, TypeScript, JavaScript, Java, SQL
Backend: Flask, FastAPI, Node.js, REST APIs
Frontend: React, Next.js
Data: PostgreSQL, MySQL, MongoDB, ETL Pipelines
AI/ML: PyTorch, Transformers, Computer Vision, NLP, LLM Systems
Delivery: Docker, Kubernetes, CI/CD, Git, Linux, Jenkins, AWS, Azure, Agile

Projects

CaptionCraft - Image Caption Generator Live | *Computer Vision, Python*
- Benchmarked transformer vision models including BLIP, GIT, and ViT-GPT-2 on 10,000 images and 100,000 captions, improving caption accuracy 15% while building Python Flask inference APIs for real-time generation at scale.

PiXMod Live | *Computer Vision, Product*
- Designed a Next.js image platform using CLIP, SAM, and Stable Diffusion, processing 1,000+ edited assets with MongoDB storage and Stripe subscriptions for computer vision workflows at scale.

CPU-scheduling Simulator Live | *C++, Systems*
- Implemented C++ scheduling algorithms including FCFS, SJF, SRT, and Round Robin with React controls and Gantt visualization, reducing average turnaround time 25% across 1,000+ simulated process combinations.